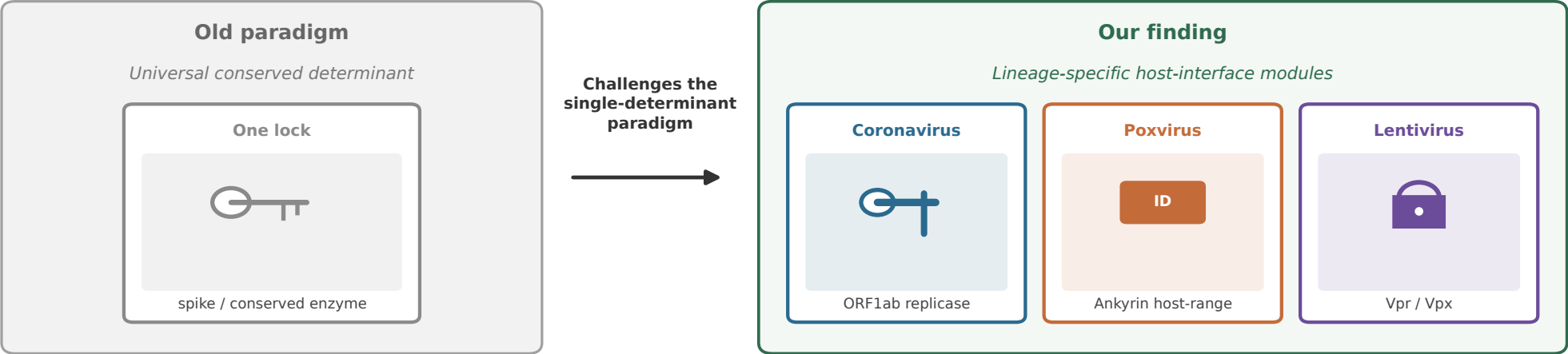


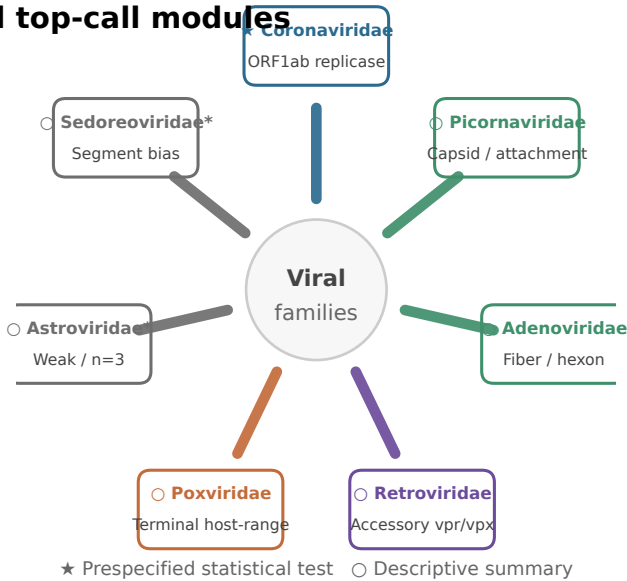
Attribution maps lineage-specific host-interface modules

Not a shared pan-viral locus — family-typical virus–host interface biology

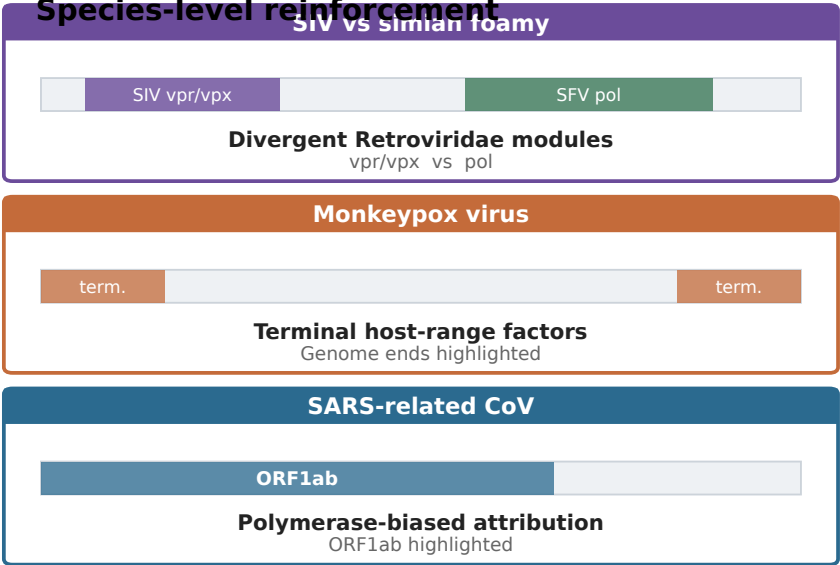
a Conceptual contrast



b Family-level top-call modules



c Species-level reinforcement



Module color key (paper-wide)



Annotation-derived interpretive classes; not causal determinants. Except Coronaviridae primary enrichment, family/species rows are descriptive top-call summaries.